

Stand-up, PPT Draft, Decision Log, Technical Understanding

1. Stand-up / Blocker Check

Dev: Not stuck. Repo review found a real (fixable) methodological issue — merging 6 years of data into one continuous timeline would have broken CTBI's day-to-day dependency at year boundaries. Resolved: Dev will use the full 6-year data for the NCTL baseline (stronger than a 1-week baseline), and reset CTBI per-year for the actual multi-day demonstration graph.

Khushi: Delivered DayStress/NCTL windows, baseline (36.34 UTCI, calculated on real data), and Risk Rubric v1.0 (5-level, with operational Low/High definitions). Day 1's data-availability note was also retroactively completed — confirmed AMC's ward-wise Census download exists, flagged elderly% and outdoor-worker data as unconfirmed.

Shweta: Validated Dev's full UTCI output — zero data quality issues found (no impossible values, no missing data, timestamps clean). Produced the first two real graphs (May 2024 hourly UTCI, 6-year daily-max overview).

Garima: 48 real ward boundaries loaded and validated (all geometrically valid). Demographic join schema built — but the actual AMC ward-wise numbers are blocked by the site's robots.txt, so this still needs a manual download by a team member. **This is the current #1 open blocker** heading into Day 3.

Vrinda: Dashboard architecture locked — Map → Ward click → Detail card now genuinely works both via sidebar and direct map click (verified the Streamlit `on_select` feature is supported in the installed version).

2. PPT — First Draft (Problem + Solution + Methodology)

Built directly into the **official SIH 2026 idea-submission template** found in the repo (docs/SIH2026-IDEA-Presentation-Format.pptx) rather than a freeform deck — this template is mandatory (max 6 slides, PDF-only submission, fixed section structure per its own instructions slide).

Filled in:

- Slide 1 (Title): PS ID 26083, full PS title, Theme (Disaster Management), Category (Software), placeholder for Team ID/Name
- Slide 2 (Idea/Proposed Solution): what the system does, how it addresses the problem, what's novel (NCTL + CTBI)
- Slide 3 (Technical Approach): real tech stack, real methodology pipeline

Not yet done (comes from Day 3-4 work): Slides 4-6 (Feasibility & Viability, Impact & Benefits, Research & References) — these depend on Garima's spatial/vulnerability results and Khushi's action engine, which aren't finalized yet.

Action needed from team: confirm actual registered Team ID and Team Name (currently assumed "TechForge" based on the repo name — please verify this matches the SIH portal registration exactly).

File: PS26083_Idea_Draft.pptx

3. Research Sources Organized

- AMC ward structure (7 zones, 48 wards) — confirmed via official AMC sources
- Ward boundary GeoJSON — DataMeet India (github.com/datameet/Municipal_Spatial_Data), CC BY 4.0
- Ward-wise Census 2011 download — confirmed to exist at ahmedabadcity.gov.in/SP/Demographics (manual download required)
- Heat Resilience Framework (Mumbai/Thane/Nagpur) — verified real, used as the conceptual basis for the 3-dimensional risk rubric
- 90th-percentile heatwave threshold method — widely used in South Asian heatwave literature (general citation, not tied to one specific unverified paper)

4. Decision Log (Master, Consolidated)

Decision	Choice	Why	Status
City	Ahmedabad	Existing Heat Action Plan since 2013, ward data feasibility confirmed	● Locked
Ward boundaries	DataMeet GeoJSON, 48 wards	Official AMC portal not easily scriptable; this source verified working	● Locked

Daytime window	06:00–18:00, mean UTCI	Captures full solar exposure, stable vs. single-hour spikes	● Locked
Nighttime window	20:00–06:00	Matches regional nocturnal-heatwave definitions	● Locked
NCTL baseline	90th percentile, full 6-year nighttime UTCI	Reproducible, climatologically grounded, not arbitrary	● Locked
CTBI computation	Per-year reset (not one continuous multi-year run)	Prevents false day-to-day continuity across year boundaries	● Locked (Dev implementing Day 3)
Risk Rubric	3D matrix (Thermal Burden × Vulnerability × Response Capacity), 5 priority levels	Matches Heat Resilience Framework + WHO/WMO graduated-response approach	● Locked
Dashboard	Streamlit + Plotly	Fastest to build given team's Python-heavy skillset; map-click interaction confirmed working	● Locked
Demographic data	AMC Census ward-wise file	Confirmed to exist; blocked by robots.txt, needs manual download	● Open — action needed
PPT format	Official SIH 2026 template (6 slides max, PDF only)	Mandatory per organizer instructions found in repo	● Locked

5. My Own Understanding of Shweta's UTCI Graph (for Day 5 Q&A readiness)

Writing this out in plain language now, so it's ready to explain to a judge without notes:

"This graph shows hourly UTCI for Ahmedabad through May 2024. Each day has a clear up-and-down cycle — UTCI rises through the morning, peaks in the afternoon, and drops at night. What matters most for our project isn't the daytime peak, which conventional heat alerts already track — it's the pattern in the troughs. Early in May, nighttime UTCI drops to around 20-25°C. By late May, those same nighttime lows only fall to about 28-30°C. That rising nighttime floor is exactly what our NCTL metric is designed to capture: the body isn't just facing hot days, it's increasingly not getting a cool night to recover. The dashed line at

46°C marks the 'extreme heat stress' threshold — by mid-to-late May, we're crossing it almost daily."

This same explanation extends naturally into introducing NCTL/CTBI without needing new slides.

Handoff

- PS26083_Idea_Draft.pptx → team, for review before Day 3-4 slides get added
- Decision Log → keep updating as Day 3 decisions get made (especially Dev's per-year CTBI implementation details)
- **Immediate action for the team: someone needs to manually download the AMC ward-wise Census file before Garima's Day 3 spatial integration.**